

Banking Loan Risk Analysis Dashboard Report

This report presents a detailed analysis of banking loan customer behavior using Microsoft Excel. The project focused on identifying repayment patterns, customer risk distribution, income segmentation, demographic lending behavior, and financial performance insights. The dashboard was developed using Pivot Tables, Pivot Charts, slicers, advanced formulas, conditional formatting, and interactive reporting techniques.

Project Name	Banking Loan Risk Analysis Dashboard
Tools Used	Microsoft Excel, Pivot Tables, Pivot Charts, Slicers
Dataset Size	20,000 Customer Records
Project Type	Interactive Financial Analytics Dashboard
Prepared By	Aakash Verma

1. Data Cleaning and Validation

The dataset underwent validation and preprocessing before analysis. The validation process ensured high-quality financial reporting and accurate dashboard outputs. Additional helper columns were created to improve segmentation and business analysis.

Validation Check	Result
Missing Income Values	0
Invalid Credit Scores	0
Negative Loan Amounts	0
Missing Employment Status	0
Invalid Interest Rates	0
Validation Status	Passed

Helper Columns Created

- Income Category
- Credit Category
- Risk Level
- Loan Size
- Age Group
- Repayment Status

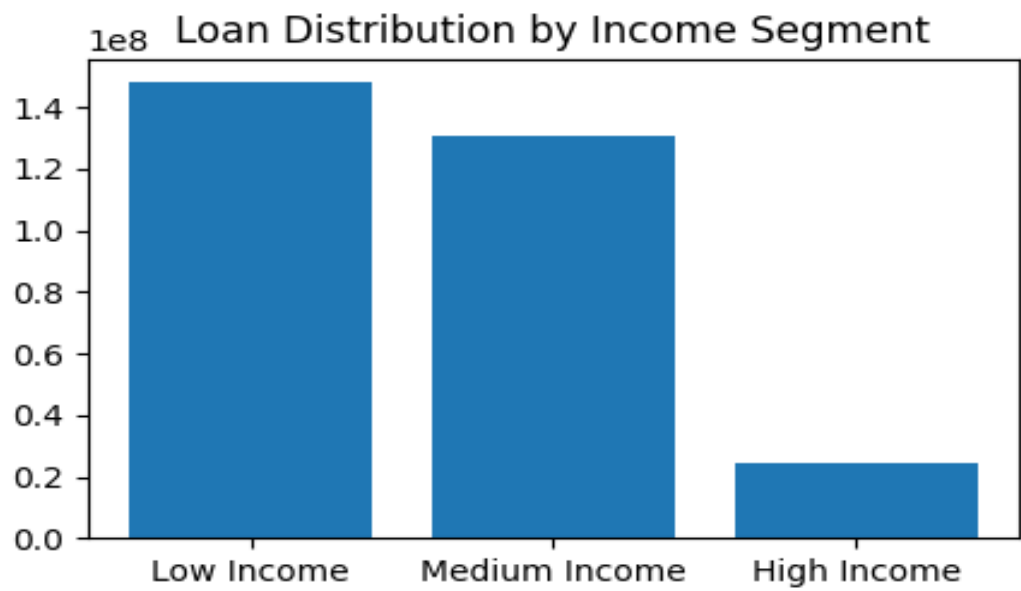
2. KPI and Dashboard Metrics

Metric	Value
Total Customers	20,000
Average Loan Amount	■ 15,129.30
Repayment Rate	80%
Average Credit Score	679.26
High Risk Customers	96

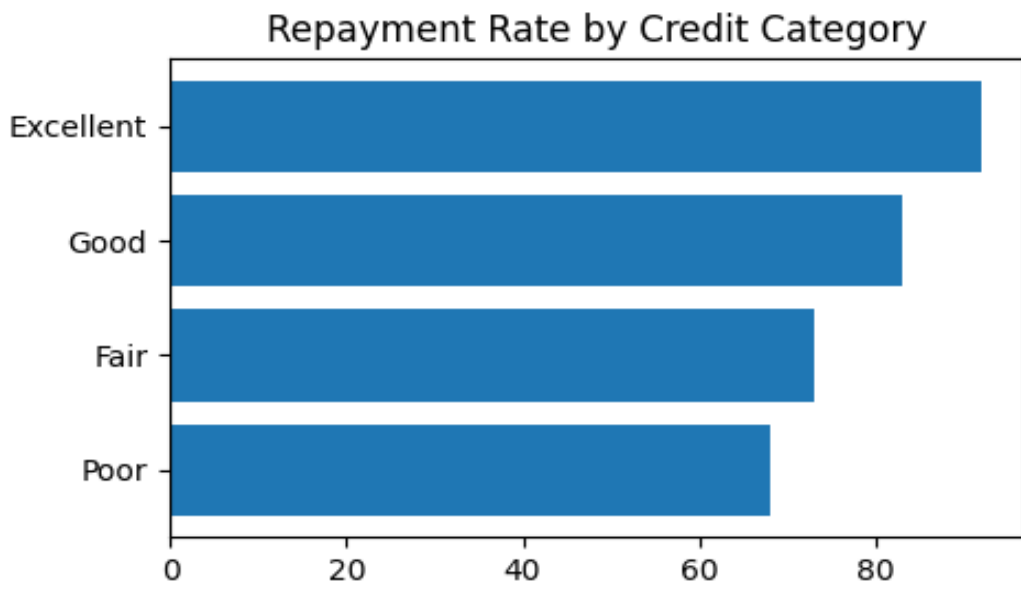
The dashboard analysis identified strong repayment performance across the loan portfolio. Excellent credit customers demonstrated the highest repayment success, while medium-risk customers formed the largest customer segment. The analysis also revealed that higher-income applicants generally received larger loan amounts.

3. Dashboard Visualizations

Loan Distribution by Income Segment

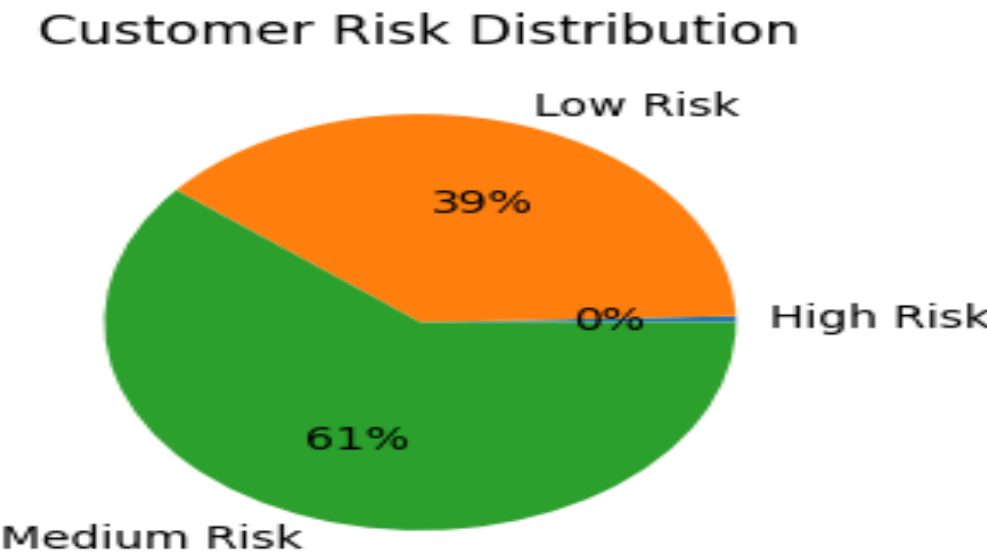


Repayment Rate by Credit Category

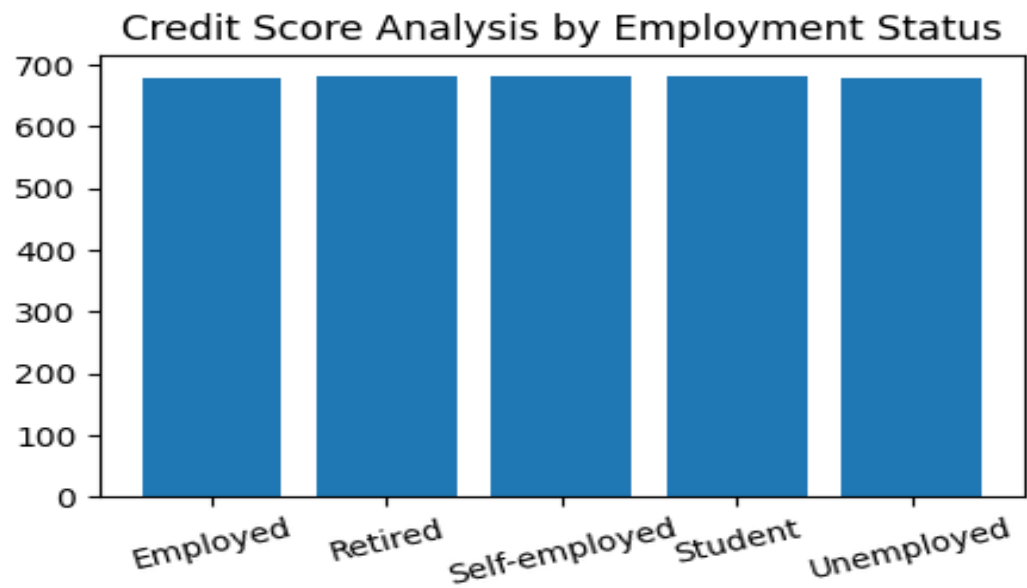


4. Risk and Customer Analysis

Customer Risk Distribution

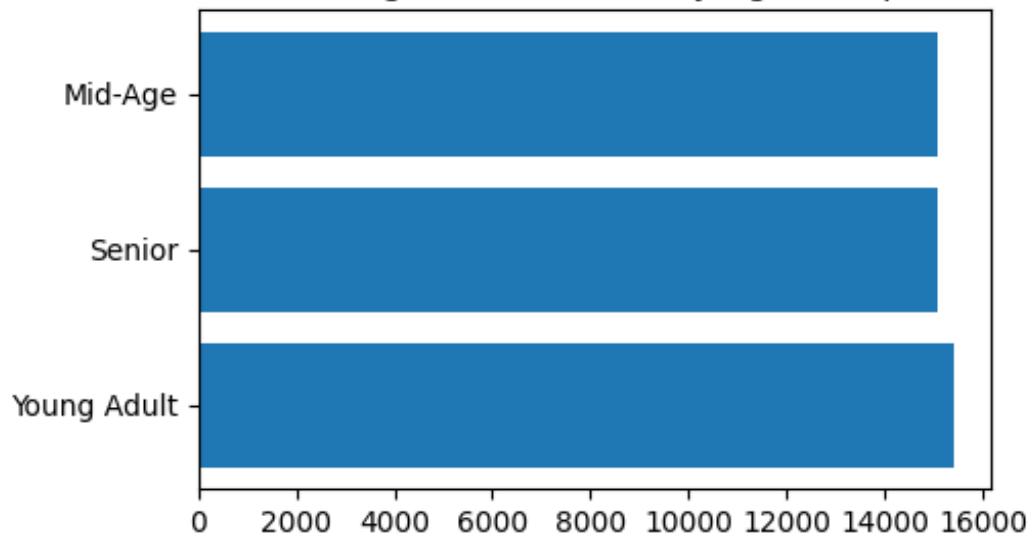


Credit Score Analysis by Employment Status



Average Loan Amount by Age Group

Average Loan Amount by Age Group



5. Business Insights and Recommendations

- Excellent credit customers achieved the highest repayment performance.
- Medium-risk customers accounted for the majority of the portfolio.
- High-income customers received larger average loan approvals.
- Poor credit customers demonstrated lower repayment performance.
- Employment status showed moderate impact on customer credit behavior.

Recommendations

- Increase lending focus on high-credit customers.
- Apply stricter monitoring for medium-risk borrowers.
- Use income and credit segmentation for smarter loan approval strategies.
- Implement automated risk-scoring systems for future banking analytics.
- Expand dashboard reporting for real-time monitoring and decision-making.

Conclusion

The Banking Loan Risk Analysis Dashboard successfully transformed raw loan data into meaningful business insights. The project demonstrated practical Excel analytics skills including data cleaning, feature engineering, pivot table analysis, dashboard design, and business reporting. The dashboard supports data-driven financial decision-making and provides valuable insights into customer repayment and risk behavior.

Prepared by Aakash Verma | Excel Financial Analytics Project